prompts is primarily statistical variation. There is some signal for BLOOM in the diagnostic (Ax-b and Ax-g) datasets. The exception is the T0 model, which shows strong performance. However, this model is finetuned in the multitask setting (similar to BLOOMZ, see Section 4.7) in order to improve performance in zero-shot prompting settings, and thus is not directly comparable to the other models shown here.

As models go from zero-shot to one-shot, variability is reduced across all prompts and models and performance slightly and inconsistently increases. Notably, BLOOM sees more of an increase in performance than comparable models when going from zero-shot to one-shot, as it is generally behind OPT in the zero-shot setting but matches or improves on it in the one-shot setting, even though it has only partly been trained on English. This may be because a multilingual language model gains more certainty in the language of input and output with a longer context.

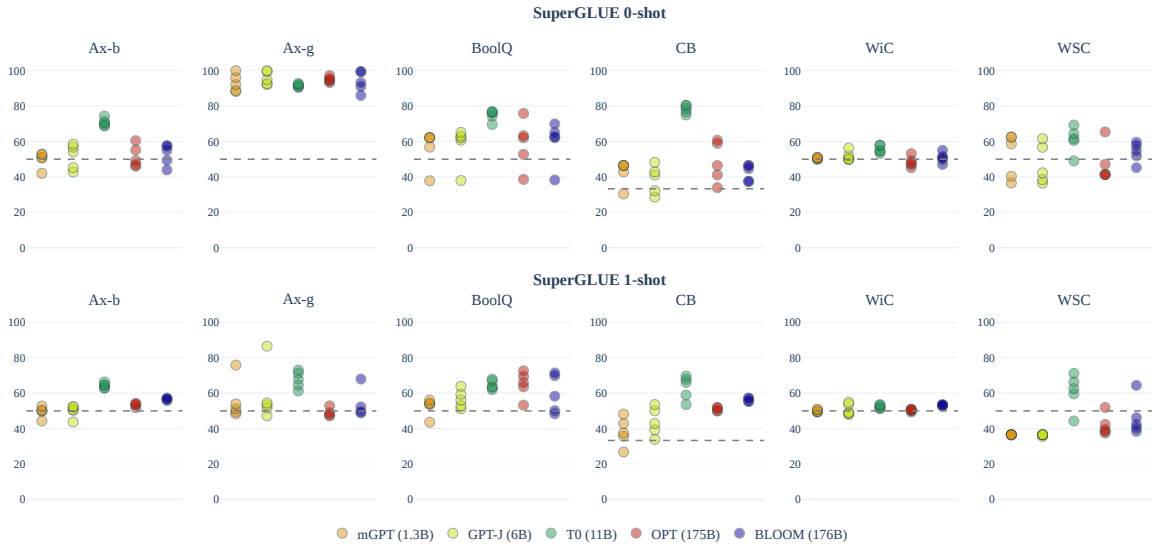

Figure 7: Performance of various LLMs on subset of tasks from SuperGLUE benchmark in zero- and one-shot prompt-based setting.

We perform an additional analysis comparing BLOOM models across model sizes. As a baseline, we also measure the average one-shot accuracy of OPT models of similar sizes (350M parameters to 175B parameters).[32] Figure 8 shows the accuracy of each prompt on each task across model scales. Both OPT and BLOOM model families improve very slightly with scale, with only models over 2 billion parameters showing signal, and there is no consistent difference between families across all tasks. In the 1-shot setting, BLOOM-176B is ahead of OPT-175B on Ax-b, CB, WSC and WiC, and matches it on the other tasks, suggesting that multilinguality does not limit performance of BLOOM on English-only tasks in the zero-shot setting.

32. We do not evaluate OPT-66B because of the lack of a similarly-sized BLOOM model.

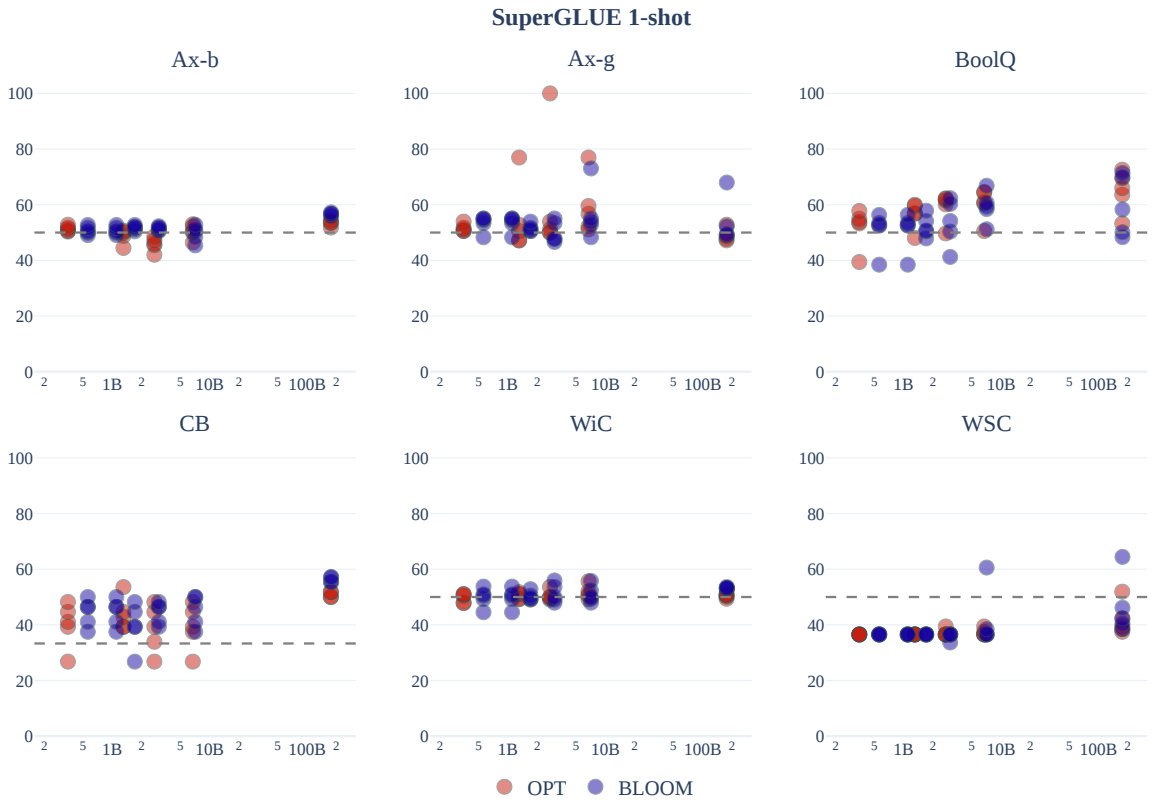

Figure 8: Comparison of the scaling of BLOOM versus OPT on each SuperGLUE one-shot task. Each point represents the average accuracy of a model within the BLOOM or OPT family of models on one of the five task prompts. The number of parameters on the x-axis is presented in log-scale.

4.3 Machine Translation

In addition to the results presented here, a more detailed analysis of BLOOM's MT quality can be found in (Bawden and Yvon, 2023).

4.3.1 WMT

WMT results for BLOOM-176B in the zero-shot and 1-shot setting are given in Table 6. The best prompts tend to be the more verbose ones; the "version-target" prompt is consistently better and the "gpt3-target" and "xglm-source+target" prompts have very poor performance, especially for zero-shot. In the one-shot setting, BLOOM can, with the right prompt, perform competent translation, although it is behind dedicated (supervised) models such as M2M-100 (43.8 BLEU for English→French and 40.4 for French→English, compared to 34.2 and 35.4 BLEU for BLOOM). The two major problems observed, particularly in the zero-shot setting, are (i) over-generation and (ii) not producing the correct language (an

obvious prerequisite for a good translation). Both of these aspects are greatly improved as the number of few-shot examples is increased.

Prompt	en → fr		fr → en		en → hi		hi → en	
Shots	0	1	0	1	0	1	0	1
a_good_translation-source+target	15.38	36.39	14.15	36.56	1.90	14.49	10.19	24.60
gpt3-target	7.90	32.55	12.73	33.14	0.26	6.51	0.66	9.98
version-target	21.96	34.22	26.79	35.42	1.96	13.95	11.48	25.80
xglm-source+target	14.91	27.83	15.52	34.51	6.80	13.62	12.05	25.04

Table 6: WMT'14 zero- and one-shot results (BLEU) for BLOOM-176B. The prompts used are described in Table 5.

4.3.2 DiaBLa

		en→fr		fr→en	
1-shot context	Truncate	BLEU	COMET	BLEU	COMET
Rand.	×	5.7	0.342	12.1	0.614
	✓	37.6	**0.634**	41.4	**0.757**
Prev.	×	6.1	0.328	12.3	0.617
	✓	**38.5**	0.614	**41.6**	0.751

Table 7: DiaBLa 1-shot results (BLEU) for the "xglm-source+target" prompt when using the previous or a random sentence as the 1-shot example (with and without truncation of outputs). In bold the best results for each direction.

Table 7 shows results testing the use of linguistic context with DiaBLa, a parallel dataset of informal bilingual dialogues. In a 1-shot context and using the "xglm-source+target" prompt, we compare the effect of using a random test set example as the 1-shot example versus using the previous dialogue utterance. In light of the overgeneration issues seen and in order to evaluate the quality of the prediction independently of overgeneration, we report results for both original outputs and after applying a custom truncation function.[33] The automatic results are inconclusive, with little difference between scores (BLEU scores are higher for previous context but COMET scores are lower). Despite these results, there is evidence in the predictions themselves that the model is able to use the context of the 1-shot example to make translation choices. See (Bawden and Yvon, 2023) for examples and further analysis.